5-1

```
create table test5_01 (First_name varchar(4) ,frequency numeric(4))
create view    tempname as(
   select substr(name ,2, length(name) -1 )as tname
   from pub.student
)
insert into test5_01(select tname First_name, count(*) frequency
from tempname
group by tname)
```

5-2

```
create view test5_02view as(
   select substr(name,2) name
   from pub.student
)
create view temp1 as(
   select substr(name,1,1) name
   from test5_02view
)
create view temp2 as(
   select substr(name,2,1) name
   from test5_02view
)
create view temp3 as(
   select substr(name,3,1) name
   from test5_02view
)
create table test5_02 as(
   select name letter, count(*)frequency
   from (
   (select name from temp1)
   union all (select name from temp2)
   union all (select name from temp3)
   ) a
   group by a.name
)
delete from test5_02
where letter is null
```

5-3

```
create view test5_03view as(
   select a.sid, b.credit
   from pub.student a, pub.course b, pub.student_course c
   where a.sid = c.sid and b.cid = c.cid and c.score >=60
   )
create view test5_03view2 as(
```

```
   select a.sid
   from test5_03view a
   group by a.sid
   having sum(credit) >= 10
   )
create table test5_03(Dname varchar(30), class varchar(10), P_count1 int ,P_count2 int, P_count int)
insert into test5_03(Dname, class )
select a.dname, a.class
from pub.student a
group by a.dname, a.class
update test5_03
set P_count1 = (
   select count(*)
   from pub.student a, test5_03view2 b
   where a.sid = b.sid and a.dname = test5_03.dname and a.class = test5_03.class
   group by a.dname, a.class
)
update test5_03
set p_count =
(select count(*)
   from pub.student
   where dname = test5_03.dname and class = test5_03.class
   group by dname, class
   )
update test5_03
set p_count1 = case
when p_count1 is null then 0
else p_count1
end
delete from test5_03
where dname is null or class is null
update test5_03
set p_count2 = p_count - p_count2
5-4
create view test5_04view as(
   select a.sid, a.class, b.credit
   from pub.student a,    pub.course b, pub.student_course c
   where a.sid = c.sid and b.cid = c.cid and c.score >=60
   )
create view test5_04view2 as(
   (select sid
   from test5_04view
   where class <=2008
```

```
    group by sid
    having count(credit) >=8
) union ( select sid
    from test5_04view a
    where class>= 2009
    group by sid
    having count(credit) >=10
))
create table test5_04(dname varchar(30), class varchar(10),P_count1 int, P_count2 int ,P_count
int)
insert into test5_04(dname,class)
(
    select dname, class
    from pub.student
    group by dname ,class
)
delete from test5_04
where dname is null or class is    null
update test5_04 a
set p_count = (
    select count(*)
    from pub.student b
    where b.dname =a.dname and a.class = b.class
    group by b.dname, b.class
)
update test5_04 a
set p_count1 = case
when exists(
        select count(*)
    from pub.student b, test5_04view2 c
    where b.sid = c.sid and a.dname = b.dname and a.class =b.class
    group by b.dname ,b.class
)
    then
    (
    select count(*)
    from pub.student b, test5_04view2 c
    where b.sid = c.sid and a.dname = b.dname and a.class =b.class
    group by b.dname ,b.class
)
else 0
end
update test5_04
set p_count2 = p_count - p_count1
```

5-5

```
create view test5_05    as(
select dname ,(
   select round(avg(maxscore),0)
   from (
   select max(score) maxscore,dname
   from pub.student d, pub.course e, pub.student_course f
   where d.sid = f.sid and e.cid = f.cid and e.name ='数据结构'
   group by d.sid, dname
   ) temp
   where temp.dname = a.dname
)avg_ds_score,
   (
   select round(avg(maxscore),0)
   from (
   select max(score) maxscore, dname
   from pub.student d, pub.course e, pub.student_course f
   where d.sid = f.sid and e.cid = f.cid and e.name ='操作系统'
   group by d.sid,dname
   )temp
   where temp.dname = a.dname
) avg_os_score
from pub.student a, pub.course b, pub.student_course c
where a.sid = c.sid and b.cid = c.cid and a.dname is not null
group by a.dname )
```

5-6

```
create view test5_06 as(
   select a.sid, a.name ,a.dname, (
   select max(d.score)
   from pub.student_course d ,pub.course e
   where d.sid = a.sid and d.cid = e.cid and e.name = '数据结构'
   )ds_score,
   (select max(d.score)
   from pub.student_course d ,pub.course e
   where d.sid = a.sid and d.cid =e.cid and e.name = '操作系统'
   )os_score
   from pub.student a, pub.course b, pub.student_course c
   where a.sid = c.sid and b.cid = c.cid and a.dname = '计算机科学与技术学院'
   and a.sid in( select sid from pub.student_course f, pub.course g where f.cid = g.cid and
g.name = '数据结构')
   and a.sid in(select sid from pub.student_course f, pub.course g where c.cid = g.cid and
g.name = '操作系统')
```

```
)

5-7
create or replace view test5_07 as(
   select a.sid, a.name ,a.dname, (
   select max(d.score)
   from pub.student_course d ,pub.course e
   where d.sid = a.sid and d.cid = e.cid and e.name = '数据结构'
   )ds_score,
   (select max(d.score)
   from pub.student_course d ,pub.course e
   where d.sid = a.sid and d.cid =e.cid and e.name = '操作系统'
   )os_score
   from pub.student a, pub.course b, pub.student_course c
   where a.sid = c.sid and b.cid = c.cid and a.dname = '计算机科学与技术学院'
   and( a.sid in( select sid from pub.student_course f, pub.course g where f.cid = g.cid and g.name = '数据结构')
   or a.sid in(select sid from pub.student_course f, pub.course g where c.cid = g.cid and g.name = '操作系统'))
)

5-8
create or replace view test5_08 as(
   select distinct a.sid, a.name ,a.dname, (
   select max(d.score)
   from pub.student_course d ,pub.course e
   where d.sid = a.sid and d.cid = e.cid and e.name = '数据结构'
   )ds_score,
   (select max(d.score)
   from pub.student_course d ,pub.course e
   where d.sid = a.sid and d.cid =e.cid and e.name = '操作系统'
   )os_score
   from pub.student a
   where    a.dname = '计算机科学与技术学院'
   )

5-9
create view test5_09 as(
   select distinct score, (
   select count(*)
   from pub.student_course b
   where a.score = b.score
   group by b.score
   ) Count1,
```

```
   (select count(*)
   from pub.student_course b
   where b.score >= 60
   )Count0,
round(
   (
   select count(*)
   from pub.student_course b
   where a.score = b.score
   group by b.score
   ) /
   (select count(*)
   from pub.student_course b
   where b.score >= 60
   )*100,2) percentage
   from pub.student_course a
   where score >= 60
   )

5-10
create view test5_10 as(
   select distinct a1.cid, (name)cname,to_char(trunc(score,-1),'fm000')||'-
'||to_char((trunc(score,-1)+9),'fm000')score,
   count(*)count1,
   (select count(*)
   from pub.course b1, pub.student_course b2
   where b1.cid = b2.cid and a1.cid = b1.cid and a1.name = b1.name and b2.score >=60 and
b2.score<=149)count0,
   round(count(*)/(select count(*)
   from pub.course b1, pub.student_course b2
   where b1.cid = b2.cid and a1.cid = b1.cid and a1.name = b1.name and b2.score >=60 and
b2.score<=149)*100,2)percentage
   from pub.course a1, pub.student_course a2
where a1.cid = a2.cid and score>=60 and score <=149 and score is not null and score not
like'% %'
group by a1.cid, name, trunc(score,-1)
)
```